

TrES-2b

R-2497

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_120829 · created 2026-08-24T22:48:32+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456168.83417 +/- 0.00017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.017
Transit depth (Rp/Rs)^2	2.21 +/- 0.51 [%]
Orbital Inclination (inc)	83.4 +/- 0.62
Airmass coefficient 1 (a1)	1.361 +/- 0.017
Airmass coefficient 2 (a2)	-0.1773 +/- 0.0044
Scatter in the residuals of the lightcurve fit is	1.26 %
Best Comparison Star	None
Optimal Aperture	4.54
Optimal Annulus	8.36
Transit Duration (day)	0.068 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.149 $\pm$ 0.017 vs 0.1254 (deviation 0.99 σ)
Duration fit vs published	0.068 d vs 0.0746 d (deviation 0.34 σ)
Mid-time O–C	-0.4 min
Depth SNR	6.3
Residual scatter	1.26 %

Light curve

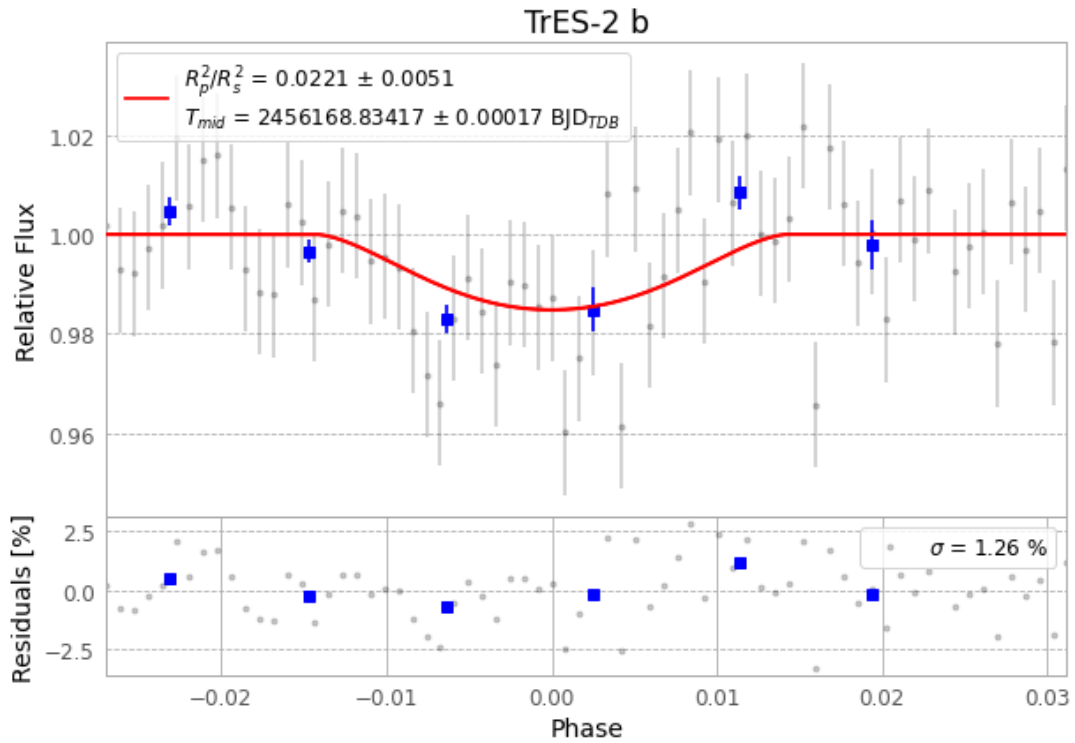


Figure 1 — FinalLightCurve_TrES-2 b_2012-08-28.png (SHA-256 dd1ecd93a079dfdd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [232, 260], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 76c94f16d93d623e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

70 FITS frames (46 MB), TRES-2120829062112.FITS → TRES-2120829094812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-120829.log (eb3b32d5c953...), FinalLightCurve_TrES-2 b_2012-08-28.png (dd1ecd93a079...), FinalLightCurve_TrES-2 b_2012-08-28.csv (150b2eb2736f...)

Compute resources

Wall time 97.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 22:48Z.